

EXPERIMENT – 4

Aim:

To identify and correct the errors in the given algorithm and flowchart for multiplying two integers without using the * operator, while achieving the minimum number of additions.

Mistakes in Given Algorithm :

- 1. Negative numbers are not handled correctly.**
- 2. Sign of the product is not considered.**
- 3. Zero cases are not explicitly handled.**
- 4. It always adds num1 num2 times.**
- 5. It does not ensure minimum additions.**

Corrected Algorithm :

- **Start.**
- **Input num1 and num2.**
- **If either number is 0, print product as 0 and stop.**
- **Determine the sign of the product.**
- **Take absolute values of both numbers.**
- **Set the larger number as addend and smaller number as count.**
- **Set prod = 0.**
- **Add addend to prod, count times.**
- **Apply the required sign.**
- **Print prod.**
- **Stop.**

Mistakes in Given Flowchart :

- 1. Negative numbers are not handled.**
- 2. There is no sign check.**
- 3. Zero is not explicitly checked.**
- 4. num1 is always used for addition.**
- 5. num2 is always used as repetition count.**
- 6. Minimum additions are not ensured.**

Corrected Flowchart :

START



Input num1, num2



◇ Any number = 0? ◇

→ Yes → Product = 0 → Print → STOP

→ No ↓

Determine sign



Take |num1| and |num2|



◇ Which is larger? ◇



Larger = addend, Smaller = count



prod = 0, index = 1

↓

◇ index ≤ count? ◇

→ Yes → prod = prod + addend → index++ → return to condition

→ No ↓

Apply sign

↓

Print product

↓

STOP

C Code :

```
#include < stdio.h >
```

```
int main( )
```

```
{
```

```
    int n1, n2, a, b, addend, count;
```

```
    int prod = 0, i, negative = 0;
```

```
    scanf ("%d %d", &n1, &n2);
```

```
    if(n1 == 0 || n2 == 0) {
```

```
        printf ("Product = 0");
```

```
        return 0;
```

```
    }
```

```
    if((n1 < 0 && n2 > 0) || (n1 > 0 && n2 < 0))
```

```
        negative = 1;
```

```

a = n1 < 0 ? -n1 : n1;
b = n2 < 0 ? -n2 : n2;

if(a >= b) {
    addend = a;
    count = b;
} else {
    addend = b;
    count = a;
}
for(i = 1; i <= count; i++)
    prod = prod + addend;

if(negative)
    prod = -prod;

printf ("Product = %d", prod);

return 0;

```

Verification:

Inputs	Product	Min addition
12, 3	36	3
3, 12	36	3
-6, 5	30	5
6, -5	-30	5
-9, 4	36	4
-4, 9	36	4
0, 8	0	0
8, 0	0	0
0, -7	0	0
-7, 0	0	0
0, 0	0	0
1, -7	-7	1

Minimum additions = $\min(|\text{num1}|, |\text{num2}|)$, or 0 if either number is zero.

Result :

The corrected algorithm and flowchart handle positive, negative and zero values correctly. All given test cases are verified, and the required minimum addition count is achieved.